

Will Kim 김기일

36-Year C++ Senior · Engine / Graphics / Physics R&D Architect · AI Agent
Harness Engineer
경력 26년 · 전 세계 1억+ 단말 배포 · 레퍼런스 대비 5~10배 성능 달성 · 코스닥
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SUMMARY

AI Agent & Runtime Engine Architect with **36+ years in C/C++** and **decades of CTO/R&D leadership**.
Shipped a focusless UI engine to **100M+ devices worldwide**; contributed to a KOSDAQ listing; consistently delivered **5–10× performance gains** over reference implementations (incl. KAIST research benchmarks). Deep cross-domain expertise spans compilers, runtimes, 2D/3D graphics rendering, real-time and particle physics simulation, SIMD, lock-free systems, and embedded — complemented by production AI agent orchestration. Currently designing and operating a production **agent harness** (*kiexpert*) implementing the orchestration layer Jensen Huang formalised at GTC Taipei 2026 as *Agent = Model + Harness*.

36 yr C/C++ (since 1990)	100M+ Devices (FXUI engine)	5–10× Perf gain vs reference impl	26 yr Software career (Lead/CTO)
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KEY ACHIEVEMENTS

GPU Shader Acceleration — 100M+ Devices Reimplemented Flash filters & 14 pixel blend modes as SIMD/SIMT GPU shaders with compile-time constant folding. Completed features Adobe left unfinished; shipped in FXUI to 100M+ devices worldwide.	Figma AutoLayout Engine (First Complete Impl.) Rebuilt WRAP/FILL/HUG layout constraints from scratch at ProtoPie. 100× integer coordinate system, child→parent propagation, bounding-box change-detection cache, compile-time constant optimisation.
Scanline Rasterizer — 5× Throughput Re-engineered ActionScript3 scanline rasterization pipeline via memory-locality & cache-friendly structures. Achieved >5× performance improvement over baseline.	ABC2CPP — Bytecode-to-Native (No JIT) Converted AS3 bytecode into optimised C++ inside dynamic libraries. Enabled iOS deployment without JIT; performance exceeded standard JIT/AOT approaches.
UE3 64-bit Migration — NetEase & Nexon Global Full UE3 + Scaleform + 3rd-party stack from 32→64-bit for NightWalker (NetEase) and NightmareBreaker (Nexon) global launches. Resolved MeshCache crash (first in-house) & Scaleform RefCount race via lock-free solution.	WkLeakDump — Zero-Source-Change Leak Tracer CRT debug heap via LoadLibrary hooking; traces mid-exit leaks; emits 2 clickable VS source locations per leak. Detected V8 engine leaks in closed-source HTS Win32 client at function-level granularity.
WkBitwiseSortSet — Novel Branchless Algorithm Cache-optimised binary tree traversal & sort; scalable set operations over millions of nodes. Full whitepaper in Portfolio PDF .	AI Agent Harness — 7,700-Line Guard Engine Production harness via Claude Code PreToolUse (~7,700 lines, 176-case self-test). Evidence-based recursive model-tier routing; gamified knowledge score; hippocampus living-memory organ (recall + foresight).

이든크루 (Edencroo) — Freelance

2025.11 – 2026.04

Senior Freelance Engineer — HTS Security, Leak Analysis & QA Automation

- **WkLeakDump**: LoadLibrary-hook CRT-dbg interceptor; zero source change; detects V8 engine leaks in closed-source HTS Win32 at function-level granularity; outputs 2 clickable allocation-site source locations in VS
- **Secure keypad**: all target platforms; full financial-sector security audit compliance
- **KCMVP pipeline**: Korea Cryptographic Module Validation Program end-to-end automation (data collection → scheduling → QA → Q&A prep → evidence packaging)
- **Boot-up QA + 수백TC QA**: startup regression + hundreds of branching test cases auto-generated & executed; WKAppBot SDK + AI for 3-exchange HTS regression without API or test hooks
- Architecture via GPT + Gemini + Claude triad synthesis (정반합 problem decomposition)

스튜디오씨드코리아 (StudioSeed Korea)

2023.06 – 2025.07

Senior R&D Engineer — ProtoPie Multi-Platform Runtime Engine

- **Figma AutoLayout engine**: first complete in-house implementation; WRAP/FILL/HUG modes, child→parent constraint propagation, 100× integer coordinate system, bounding-box change-detection cache
- **Android rendering (PPPaint)**: object pooling + eliminate redundant setColor calls; Android Studio Hotspot profiling; removed freeze root causes; improved overall engine stability
- **TDD architecture**: 300+ unit tests; algorithms applied directly to real-time UI engine frame stabilisation & layout performance
- Env: MacBook M2 Pro, Windows x64 / KMM (Kotlin Multi-Platform), UE 5.2/5.4, Android Studio, Xcode

원더피플 (Wonderpeople) — 에이스툼 납품

2019.03 – 2023.02

Engine Architect — NightWalker (NetEase) & NightmareBreaker (Nexon) UE3 Migration

- **Full 64-bit migration** of UE3 + Scaleform 4.0/4.1 + all 3rd-party code; resolved every derived issue
- UE3 Serialize buffer→compile-time constant optimisation; GC Mark 5+ branch-type optimisation
- **Scaleform RefCount race** (game/render threads): eliminated via lock-free solution (first in-house)
- **MeshCache crash**: deep-trace multi-week analysis → complete stabilisation (first company-wide)
- **Game QA autobot**: actor/particle motion prediction + keyboard/mouse injection; automated quest, combat, PvP paths; collected hundreds of thousands of crash reports for proactive regression prevention

Result: NightWalker global launch — nb.163.com / nightwalker.nexon.com

피아소프트 (Piai Soft)

2013.08 – 2018.10

CTO — FXUI Engine & Runtime R&D

- **FXUI renderer**: multi-core + OpenGL ES2 batching; GPU BitmapData acceleration; **10 GPU bitmap filters + 14 pixel blend modes** at full spec (completed features Adobe left unfinished)
- **AS3 JIT compiler**: SIMD machine-code optimisation; ABC2CPP (AS3→C transpiler) for JIT-less iOS deployment
- **AwBlinPhongShader**: runtime optimal shader generator (env-map, light count, material combinations)
- **Naver Entry coding-education player**: 2× speed / 1/10 memory vs existing Adobe player; no content conversion required; Android/iOS/Windows/Linux
- **PoringLove game engine**: high-perf Live2D animation + GPU particle motion blur + 128-ch SoundMixer; memory 1/5 of comparable games; glyph-level Korean lip-sync
- **B2B embedded deliveries**: NTT PLALA (4K/FHD TV), SHARP (MFP), JVC Kenwood (AVN), LG (Wearable/DTV/SmartTV), LEO AIC (EV Concept Digital Cluster)

디지털아리아 / 지트리비엔티 (Digital Aria)

2007.07 – 2013.06

상무이사 (Executive Director) — FXUI Mobile Solution Engine

- **AS3 JIT compiler** from scratch; ported AS3 runtime to most mobile platforms (ARM/MIPS/Intel)
- **FXRaster**: complete ground-up redesign of software rasterizer with machine-code-layout-aware extreme optimisation
- **CasaUI Unity3D C++ plugin**: MonoVM (C#) ↔ FXUI organic binding for high-performance GUI
- Result: **100M+ devices worldwide**; B2B stack contributed to **KOSDAQ listing**

플레이위드 (Playwith) — 써니와이앤케이 납품

2003.04 – 2005.06

Lead Engineer — Game Portal & KSCRIPT Scripting Engine

- **KSCRIPT**: server-side script engine with hot-reload without restart (10 years before Node.js); ECMA JS 99%+ compatible; 3-stage optimisation; 2.5× PHP throughput; threading built in
- **LemonBall game portal**: IOCP real-time DB cache daemon, real-time chat/board without ActiveX, MSN Passport auth, multi-ISP protocol standardisation (KMS)

모비로 (Mobiro) / 셀포인트 (Cellpoint)

2002 – 2007

System Architect

- Mobiro: real-time SMS/MMS mobile gateway daemon for Korea's 3 MNOs; MOBIGATE HTML→WAP auto-converter; 3-month+ zero-downtime operation after eliminating all error-handling edge cases
- Cellpoint: online game server/engine design; IOCP server library authored & distributed

노브레이크테크놀러지스 (Nobreak Technologies)

1998.06 – 2002.03

Founder / Lead Developer

- **CrazyWWWBoard 2000**: de facto standard Korean web bulletin board; shipped to Chollian, Hitel, NowNuri and NEC/Tokyo Electric (Japan edition); embedded script engine, dynamic design switching
- CrazySearch multimedia search engine; CrazyArchive e-mail→BBS converter; qDecOOP OOP library (company-wide base); RMTMAKE remote compile automation (2× Unix dev speed)

납품 레퍼런스 (KEY CLIENT DELIVERIES)

Client	Domain	Contribution
NetEase (China)	Global MMORPG	NightWalker UE3 64-bit migration & global launch foundation
Nexon	Global MMORPG	NightmareBreaker UE3 64-bit migration & crash stabilisation
NTT PLALA (Japan)	4K/FHD Smart TV	Embedded FXUI engine delivery
SHARP (Japan)	High-End MFP	Embedded FXUI engine delivery
JVC Kenwood (Japan)	Premium AVN	Embedded FXUI engine delivery
LG Electronics	Wearable · DTV/SmartTV	Embedded FXUI engine delivery
Hyundai Motor	HMI Interaction POC	UE5 plugin: multi-layer UI event passthrough, script binding, UX orchestration
이든크루 (Edencroo)	HTS Security & QA	WkLeakDump · Secure keypad · KCMVP · Boot-up & TC QA automation (WKAppBot+AI)

WkAppBot SDK — Windows/Android UI Automation Framework for AI Agents

MIT Open-Source

v7.6

- **Focusless-first:** UIA Invoke/Value/Toggle + Win32 PostMessage for background automation without focus steal; legacy MFC/HTS via WM_CHAR
- **Self-healing (RAG):** CCA segmentation + OCR + Gemini Vision triple cross-validation; results cached in Vector DB; BLOCKER overlay detection via threaded UIA+CDP cross-validation
- **CDP-direct LLM web agents:** pure CDP/WebSocket for Gemini & ChatGPT without API keys; real-time streaming, per-session incognito contexts
- **wkdoctor suite:** 10+ health checks, emergency-kill watchdog, WMI culprit ID, session-tree sparing; Antigravity CLI (agy) + wkdoctor auto-repair of broken alias symlinks
- github.com/kiexpert/wkappbot-sdk · kiexpert.github.io/wkappbot-sdk

wkharness / kixpert — AI Agent Harness & Living Knowledge Base

Production

- ~7,700-line Claude Code PreToolUse guard engine; 176-case self-test suite; evidence-based recursive model-tier routing (Opus→Sonnet→Haiku→Codex)
- Living Knowledge Tree (T1/T2/T3 lazy-load); self-evolving skill catalog; gamified knowledge score; hippocampus living-memory organ (recall + foresight sessions)
- Token pace/budget throttling; weekly + monthly safety floors; stateless transcript-derived enforcement; autonomous system-resource watchdog

WkWave MAFT — DFT Multi-Resolution Adaptive Fourier Trader

Live Trading

- DFT 2:3:5 multi-resolution frequency analysis for market signal extraction; live orders via Hantoo REST API + Kiwoom HTS CDP/UIA automation
- Serverless 5-layer always-on (GitHub Actions cron); 565-pair stock correlation; gap-up backtested +6.73% T+1; Groq LLaMA + Gemini AI market advisor

WkBitwiseSortSet — Novel Sorting Algorithm

Portfolio PDF

- Branchless, cache-optimised binary tree traversal & sort; scalable set operations over millions of nodes
- [Portfolio PDF](#)

TECHNICAL SKILLS

Languages	C/C++ 36+ yr (expert since 1990), Python, Kotlin, Java, JavaScript, ActionScript3, x86/ARM Assembly
Graphics & Rendering	OpenGL / GLES2 / OpenCL, GPU shader programming (SIMT/SIMD), scanline rasterizer design, 2D/3D engine architecture, FBX pipeline, Live2D, Figma AutoLayout
Physics & Simulation	Real-time interactive physics, particle systems (GPU motion blur), rigid-body & physics-engine integration (UE3), DFT/FFT signal processing
Engines / Platforms	Unreal Engine 3 & 5, Unity3D (MonoVM C++ plugin), embedded (ARM/MIPS/NonOS), iOS, Android, Windows Win32/x64, Linux
Compilers & Runtimes	JIT/AOT compiler design, bytecode-to-native transpilers (ABC2CPP), DSL interpreters, VM & hybrid runtime engines, AS3 runtime, KSCRIPT
Performance	SIMD (SSE/NEON), cache-friendly structures, branchless algorithms, lock-free / IOCP, memory bandwidth optimisation, profiling (VTune, Telemetry, Android Hotspot)
AI / Agent	LLM, RAG, MCP, Vector DB, multi-agent orchestration, harness engineering, Claude / OpenAI / Gemini API, prompt engineering, CDP browser AI agents, token-cost & tier governance
Security	Secure keypad, KCMVP certification pipeline, CRT debug heap hooking (LoadLibrary intercept), leak tracing, cryptographic module validation
Tools	Visual Studio, Android Studio, Git / GitHub Actions, CDP (Chrome DevTools Protocol), Win32 UIA accessibility, TensorFlow

EDUCATION

홍익대학교 세종캠퍼스 (Hongik University, Sejong)

1993.03 – 2001.02

컴퓨터정보통신공학 (Computer Information & Communication Engineering)

Research: fractal image compression (professor collaboration, 1993). HVIEW Korean text viewer (nationwide distribution, full magazine feature, 1994–1995). Commendation (장려상) for real-time equipment monitoring system, ROKAF (1997).

EARLY PROJECTS & MILESTONES

1993	프랙탈 이론 이미지 압축	Hongik University professor research; image processing library
1994–1995	HVIEW (공개소프트웨어 한글 뷰어)	Full computer-magazine feature; nationwide public distribution
1996	DOS 싱글태스킹 멀티스레딩 엔진	Time-slice multithreading under single-tasking DOS; developed during ROKAF service
1997	장비 실시간 감시 · 전화연락 시스템	ROKAF internal; awarded 장려상
2001	CWBKMS 지식관리시스템	Async UI update 4 years before AJAX; adopted as company-wide KMS